A Joint Opportunistic Energy Harvesting and Communication System Using VLC for Battery-Less PV-equipped IoT

Khodr Hammoud, Dominique Schreurs, Sofie Pollin, and Zhuangzhuang Cui
Div. ESAT-WAVECORE
KU Leuven, Leuven, Belguim
Email: {khodr.hammoud, dominique.schreurs, sofie.pollin, zhuangzhuang.cui}@kuleuven.be

***Abstract*—The trend of high-speed Internet of Things (IoT) devices with secure, interference-free and health-neutral communication promotes Visible Light Communication (VLC) as a promising candidate to complement or replace radiofrequency communication for indoor applications. This paper proposes a Joint Opportunistic Energy Harvesting and Communication system (JOE-HC) using VLC. This system can perform Simultaneous Light Information and Power Transfer (SLIPT) to support battery-less photovoltaic (PV)-equipped IoT devices. The spatial Signal-to-Interference-plus-Noise-Ratio (SINR) is optimized using the Spatial-Time Division Switching (S-TDS) scheduling approach in the JOE-HC. In this approach, we add a time delay guard to eliminate the signal interference caused by the switching impairments of the light transmitters, where the Light-Emitting-Diode (LED) serves for illumination and communication purposes. The simulation results show that the SINR at a time slot has increased a maximum of 9 dB in the overlapping regions by employing the proposed scheduling scheme.**

***Index Terms*—Internet of Things, Photovoltaics, Light-Emitting Diode, Visible Light Communication, Battery-less, Energy Harvesting.**

I. Introduction

The Internet of Things (IoT) devices is rapidly increasing. Nowadays, there are about 30 billion connected IoT nodes. By 2023, the number of IoT nodes forecasts to double [1]. IoT devices coexist in various environments, e.g., hospitals, clinics, home health monitoring [2], etc. Due to this, desensitization, a process that weakens the receiver's sensitivity to receive the desired signal, is a common interference and noise that occurs [3]. Most of these IoT nodes use RadioFrequency (RF) technology to connect and communicate [2]. The increased pace of connected IoT devices presents new challenges for the already congested RF spectrum [4]. As an RF alternative communication, Optical Wireless Communication (OWC), specifically Visible Light Communication (VLC) for indoor applications, is gaining attention among research communities.

The VLC spectrum spans between 430 THz and 750 THz, where the data rate can reach 100 Gbps, and these bands have no electromagnetic interference with the currently deployed RF communication systems. Furthermore, the operating spectrum of VLC is unlicensed and less sensitive to congestion [4], given the significant path losses. All these offer an RF spectral relief.

In general, visible light offers two purposes: illumination and communication. For both purposes, the transmitted light can be seen as a source of energy harvested opportunistically to provide a sustainable and autonomous battery-less IoT node. Currently, the energy capacity for IoT devices is supplied by batteries, which need to be replaced periodically, resulting in an increase in labour costs and more chemical waste [5]. Light energy harvesting can provide, thanks to its high power density, energy to sustain the operation of the IoT device seamlessly. Our recent work [6] shows that if the waveform of the transmitted light is modified, a promising increase in Power Conversion Efficiency (PCE) is achievable. Hence, the IoT node, i.e. wearable sensor, can stay longer awake to do long-term monitoring and provide better Quality of Service (QoS) [7].

Although RF power transfer and communication is a mature technology and research domain, the radio spectrum's congestion is still a challenging research issue. VLC offers spectral relief for the congested RF spectrum. VLC, especially by using photovoltaic (PV) panels as receiver front-end instead of photo-diodes, attracted many researchers. In analogy to *Simultaneous Wireless Information and Power Transfer* (SWIPT), *Simultaneous Light Information and Power Transfer* (SLIPT) is accomplishable. The authors of [8] were the first to study SLIPT using a PV panel as a front-end photo-detector in an OWC system. The authors experimentally demonstrated this feasibility. In [9], the authors studied a trade-off between energy harvesting for battery-less IoT and data communication. The authors were able to achieve 1 Gbps at two meter. A circuit for SLIPT purpose was designed and measured by the authors of [10]. Even though the achievable distance (32.5 cm) was short, the paper showed a practical circuit and measurements.

On the other hand, simulation-based research showed the possibility of modeling the optical transmitter, the channel and the optical received signal. The authors in [11] presented theoretical modeling and hardware implementation for infrared wireless communication. Even though it was for infrared light and photo-diode, we can use the same modeling approach for PV-based SLIPT IoT nodes using VLC since visible light and PV obey the same physics as infrared and photo-diode.

A. Motivation

The prior studies on PV-based VLC have predominantly concentrated on the harvested energy and Bit Error Rate (BER). BER is a function of the rate and the Signal-to-Noise Ratio (SNR). These studies have adopted a single-link analysis approach, examining multi-transmitter, single-receiver IoT node configurations. To fill this gap, this study presents an approach for *Joint Opportunistic Energy Harvesting and Communication (JOE-HC)* using indoor LED modulated light.

The research examines multiple IoT node coexistence in confined spaces, such as hospital waiting rooms, where the interference among IoT nodes is high in the overlapping regions illuminated simultaneously by multiple transmitters. Consequently, we study the Singal-to-Interference-plus-Noise-Ratio (SINR) instead of SNR.

A time-scheduling approach is proposed to optimize the SINR in the overlapping transmission regions and hence the communication capacity. The framework works as follows: if an LED sends information, the other LEDs only illuminate. The opposite of the previous operation occurs at a different time slot, which means this approach utilizes a Time Division Multiple Access (TDMA) scheme. TDMA perfectly avoids interference; however, it does not consider hardware impairments. One significant impairment is the delay (δt) caused by switching hardware on and off. This delay becomes essential in dynamic environments where the position of the IoT nodes in the room is unknown.

B. Main Contributions and Paper Structure

In this paper, we propose Spatial Time Division Switching (S-TDS), an approach similar to TDMA but with an added time delay to encounter the impairment of the transmitter hardware. The workload of this approach is split among the LEDs. Since the IoT device is non-stationary and hardly communicates its positions to preserve energy, various LEDs send the exact copy of the information sequentially. Thereby, the overlap between sequential transmissions from different LEDs is avoided by taking a sufficiently long idle gap between transmissions. If no data communication is needed, all the LEDs illuminate using a Pulse Width Modulation (PWM) waveform with a predefined duty cycle optimized for energy harvesting [6].

Furthermore, we provide mathematical equations describing light propagation, available power per PV area, harvested power and the SINR, all in euclidean form. These closed-form equations are derived using a PV panel's short and open circuit conditions. Although these two conditions do not represent the actual behaviour of the PV panel, it gives enough insights to perform optimization and trade-offs. Furthermore, we evaluated the performance of S-TDS, where we achieved an SINR increase with a maximum of 9 dB in the overlapping regions.

The remainder of this paper is organized as follows. Section II outlines the related work on interference mitigation in coexisted IoT devices. We describe the system model, by which the light propagation, the energy harvesting and the SINR mathematically are illustrated in Section III. The model is simulated for different scenarios, and their results are discussed in Section IV. Finally, we conclude and propose a few action points for future work in Section V.

II. Related Work

The mitigation of interference for coexisting IoT devices is well investigated in the literature. In [12], an extension for the existing IEEE 802.15.6 scheme is addressed. The authors proposed to use a Superframe interleaving clustering of IoT devices to reduce the interference between IoT nodes. Although the throughput, package dropping, and latency were improved, the authors did not address the influence of other IoT nodes that were not in the cluster. A similar approach has been addressed by [13]. However, the authors did not explain the energy budget required for such an approach to the IoT.

The authors of [14] were aware of the intra-interference of IoT nodes. They proposed an Exchange Free Resource Scheduling scheme to alleviate interference in the dynamic coexistence of IoT devices. The authors divided the frame into data transmission and retransmission periods. However, this scheme did not consider that power for IoT devices is a scarce source.

Another approach using machine learning or artificial intelligence (AI) is illustrated in [15], [16]. Despite this type of proposal being very modern and aligning with AI's trend, it did not indicate the energy consumption of such an AI-enhanced approach. Furthermore, this approach did not show the system's complexity for both the transmitters and the receivers.

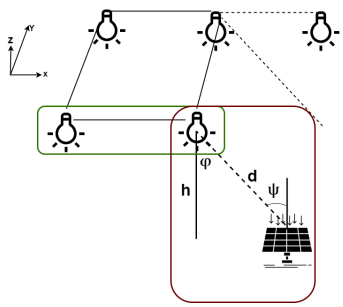
III. System Model

In an indoor VLC scenario, we consider a room with a layout of $3 \times 3 \times 3$ m. There are i LEDs (transmitters) denoted as Tx_i, distributed in the room to illuminate in two modes: illumination using a PWM waveform with a predefined duty cycle and illumination with data transmission. The latter is switched in an S-TDS pattern. The LEDs distribution and the receiver's position Rx, PV panel, are presented in Fig. 1a. We show only one receiver in Fig. 1a; however, there are potentially multiple users in the room. To generalize our study, we assume all LEDs' functions and propagation geometries are identical.

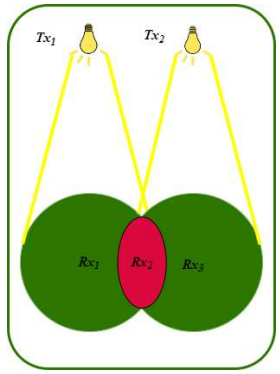
To ease the analysis of the system and for better illustration, we will use a two LEDs scenario. The system architecture of the two LEDs VLC system is presented in Fig. 1b. In such architecture, there are multi-user PV-equipped IoT nodes.

Fig. 1b illustrates the interference region where a receiver Rx_2 receives signals from both transmitters. To reduce such interference, the LEDs will operate in two modes: Illumination mode using PWM waveform ($mode\ I$) and Illumination along with communication ($mode\ II$), where the waveform is modulated to transmit the data bits, see Fig. 2a.

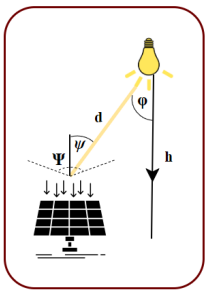
The two operation modes are allocated to different time slots to eliminate interference in the overlapping regions. Given the propagation and switching delay, some overlap between



(a) LED distribution in VLC.



(b) Two LED transmitters with multi-user PV receivers.



(c) Illustration of light propagation geometry for a typical LED.

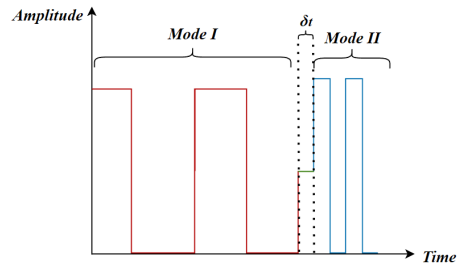
Fig. 1: A system overview consists of two LED transmitters with multiple user PV receivers (b) that demonstrates light propagation geometry using one LED (c), including LED distribution for VLC (a).

modes is possible at the receiver. While all the LEDs are in *mode I*, assume now Tx_1 is activated in $mode\ II$; once it is done with transmitting the data, a time delay is introduced to overcome the switching delay of the hardware. Meanwhile, Tx_2 operates in $mode\ I$ until the δt time slot is over. Then Tx_2 is activated to send the exact copy of the data. This makes sense mainly for broadcasting data while the position of the IoT node is unknown and non-stationary. Note that once each LED transmits the needed data packet, it returns to *mode I*. For a better illustration of time allocation, see Fig. 2b.

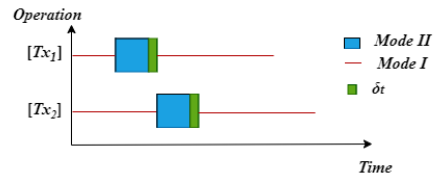
It is worth mentioning that switching on a high crest factor compared to a constant waveform enhances the energy harvesting of the PV-equipped IoT device [6]. Nevertheless, the switching should not cause flickering if it is fast for human's eye perception [17].

A. LED Light Propagation: Channel Model

The LED is placed at height (h) relative to the receiver's x and y coordinates. The LED has an irradiation angle (φ), while the receiver has an incidence angle (ψ). These angles are equal, assuming the Line-of-Sight (LOS) condition. Furthermore, the



(a) LED's operating modes.



(b) An illustration of illumination and communication using S-TDS approach.

Fig. 2: An illustration of illumination and communication using the S-TDS approach is presented, including the operating modes of LEDs.

receiver has a Field-of-View angle (FOV) noted as Ψ, see Fig. 1c.

Now, the distance (d) can be written as follows:

$$d = \sqrt{(X_{PV} - X_{T_x})^2 + (Y_{PV} - Y_{T_x})^2 + h^2}, \tag{1}$$

where X_{PV} and Y_{PV} are the x and y coordinates of the PV panel accordingly, and X_{T_x} and Y_{T_x} refer to the coordinates of the LED transmitter. Following this, φ can be calculated by

$$\varphi = \arccos\left(\frac{h}{d}\right). \tag{2}$$

The angular distribution of the radiant light intensity is modelled as Lambert's cosine law [18]. The cosine law holds that the radiant intensity is sent in all directions independent of the observation angle. The Lambertian parameter order m, which calculates the light source's radiation pattern, is dependent on the physical parameter of the light source, in particular, the Full Width at Half Maximum (FWHM), $\Phi_{\frac{1}{2}}$. In the formulation of the Lambertian parameter, we assume that the FWHM is the same as the irradiation angle. Then, we have

$$m = \frac{-\ln(2)}{\ln(\cos(\Phi_{\frac{1}{2}}))} = \frac{-\ln(2)}{\ln(\cos(\varphi))}. \tag{3}$$

Most OWC systems, including VLC systems, are strongly LOS yielded. In that sense, the horizontal illuminance (E_{hor}) can be deduced using eq. (2) and (3) as follows:

$$E_{hor} = \frac{I(0)\cos^m(\varphi)\cos(\psi)}{d^2}, \tag{4}$$

where $I(0)$ is the centre luminous intensity of the LED measured in *candelas* (cd), we notice that eq. (4) can be

reduced further. In our study, we consider a LOS condition without other multipath. Hereafter, φ and ψ are equal, hence,

$$E_{hor} = \frac{I(0)\cos^{(m+1)}(\varphi)}{d^2}. \tag{5}$$

Now, we have almost all the ingredients to describe the light propagation, including the channel loss (H_{LOS}), which is described as:

$$H_{LOS} = \begin{cases} \frac{(m+1)A_{PV}\cos^m(\varphi)\cos(\psi)}{2\pi d^2} & , 0 \leq \psi \leq \Psi \\ 0 & , \psi > \Psi \end{cases} \tag{6}$$

where A_{PV} is the area of the solar panel measured in square meters (m^2). Note that we did not include filter attenuation or concentration gain.

B. Opportunistic Energy Harvesting

The light intensity received at the PV panel depends on the area of the PV panel, the distance from the LED and any obstacles in the LOS. Furthermore, another essential factor influencing the conversion of the light intensity into energy is the responsitivity of the PV panel, which is the efficiency of light into power conversion. In simple words, the available power at the PV panel front end can be formulated as follows:

$$P_{available} = P_t H_{LOS}, \tag{7}$$

where $P_{available}$ is the available power and P_t is the optical transmitted power.

However, eq. (7) does not give insights about the PV panel parameters. For that, the generated current of the PV panel should replace the $P_{available}$ in eq. (7). We know that eq. (7) can be rewritten as a function of voltage and current. It is then expressed as:

$$P_r = \eta I_{PV} V_{oc}, \tag{8}$$

and using the short circuit condition [19], then:

$$I_{PV} = \eta H_{LOS} P_t, \tag{9}$$

where P_r is the received power, η is the filling factor of the PV panel and I_{PV} is the short circuit current. Next, the output voltage, the open circuit voltage [19], of the PV panel is expressed as follows:

$$V_{oc} = V_t \ln\left(1 + \frac{I_{PV}}{I_0}\right), \tag{10}$$

where V_t is the thermal voltage and I_0 is the dark saturation current.

Hereafter, the equation for power harvesting (P) using a PV panel is formulated using eqs. (7), (9) and (10) as:

$$\begin{aligned} P &= \eta I_{PV} V_{oc} \\ &= \eta H_{LOS} P_t V_t \ln\left(1 + \frac{I_{PV}}{I_0}\right). \end{aligned} \tag{11}$$

Eq. (11) does not consider the PV panel's responsitivity. As mentioned before, the responsitivity of a PV panel is the efficiency by which the PV panel is capable of converting light into electricity. This conversion capability depends on the wavelength of the light and the material used in producing the PV cell, which makes the analysis cumbersome. Therefore, the responsitivity is omitted from eq. (11).

TABLE I: LED specifications & Parameters

Parameter	Value
Electrical Power	2.5 W
Wall plug efficiency	40%
η	75%
A_{PV}	0.0023 m^2
h	2.5 m
FWHM	15°
Data rate	1 Mbps
δt	10 ns
V_t	20 mV
I_0	10 nA

C. SINR

Based on the equations from Sec. III-B, formulating an equation for SNR is straightforward, following the definition of SNR. It becomes as follows:

$$SNR = \frac{P}{\sigma^2}, \tag{12}$$

where σ^2 is the noise power, the dominant noises in this equation are shot and thermal noise.

However, eq. (12) does not include the interference between the LEDs. For that purpose, the interference power of one LED should be included to eq. (12) once the other LED is switched off. eq. (12) becomes as:

$$SINR = \frac{P}{\sigma^2 + \alpha P_I}, \tag{13}$$

where P_I is the interfering received power equal to eq. (11) and α is the total time delay of switching the LED off and propagation delay.

The time delay (α) is a sum of the switching delay (δt) and the propagation delay time (τ) needed for the light to reach the PV panel. τ can be calculated as:

$$\tau = \frac{d}{c}, \tag{14}$$

where d is same as in eq. (1) and c is the speed of light. Then α becomes:

$$\alpha = \delta t + \tau. \tag{15}$$

IV. Simulations and Results

To evaluate the analysis in the previous section, we simulate the system using MATLAB environment. We placed two LEDs on the room's ceiling, taking the centre of the room as a reference point. Then, the two LEDs are separated by a distance of 1.32 m to obtain a uniform illumination between the LEDs [20]. Next, the PV-equipped receiver is placed at a height of 2.5 m from the ceiling. It means that the receiver is 0.5 m above ground. Furthermore, the LED specifications and other simulation parameters are included in Table I.
Markedly, the wall plug efficiency is the capability of LED to convert electrical into optical power. Furthermore, the FWHM is the same as the irradiation angle, i.e., φ.

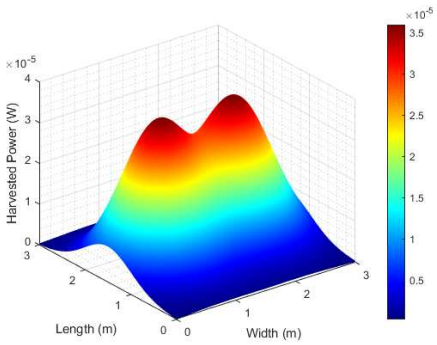


Fig. 3: The harvested power at 2.5 m using two LEDs.

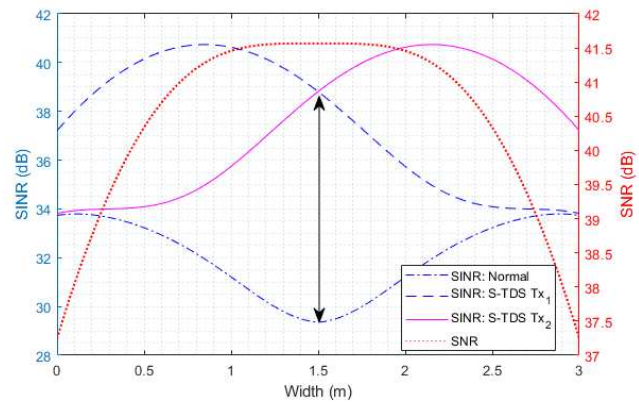


Fig. 4: Simulated Spatial SNR distribution, SINR without S-TDS approach and SINR using S-TDS approach for two LED transmitters.

A. Simulations Scenarios

This section evaluates the JOE-HC system using the parameters listed in Table I. At first, we present the results of horizontal illumination and the harvested energy using two LEDs. We will demonstrate some results in 3D plotting for better visualization and others in 2D for better reading and analysis. Next, we simulate the spatial SNR assuming there is no interference. Hence, switching the LEDs is ideal. Furthermore, we introduce the interference noise due to switching impairments from the LEDs. Finally, we evaluate our S-TDS as presented in Fig. 2b, all based on the simulated SNR or SINR.

B. Results and Discussion

The harvested power at the height of 2.5 m from the ceiling is simulated for two LEDs in $mode\ I$; see Fig. 3. The power in the figure represents the harvested optical power at the front end of the PV-equipped IoT sensor. It is noticeable that the illumination is not uniform in the whole room. This is because we placed only two LEDs in the room. It should be noted that the z-axis represents the magnitude of power.

We observe that the maximum power available or harvested is directly under the LED at $\varphi = 0$. Furthermore, the PCE, the ratio of harvested power divided by the available power, is 18%. We used the available power calculated using eq. (7) to calculate this PCE.

First, we simulate the SNR, $mode\ II$, representing the maximum achievable capacity in an ideal scenario without interference. We added only the shot and thermal noise illustrated in eq. (12) to perform the simulation. We assume that the switching is ideal and there are no impairments. The results are in Fig. 4. The SNR represents an ideal communication with no interference from one transmitter to another.

Then, the SINR for the simultaneous transmission is simulated and depicted in Fig. 4. In this scenario, the influence of switching impairments on SINR is included. The result shows that the spatial SINR distribution reduces, especially at the room's centre, compared to SNR. Also, the peaks under the LEDs are shifted spatially. This results from the interference addition of signals at the front end of the IoT node. We simulated the SINR using 10 ns, a typical rise and fall time of switching, for the δt. Note that we assumed that the δt is the same for both LEDs switching hardware. However, this does not occur in a practical situation. Each LED has its switching delay impairment. That means the interference at the room's centre shifts to the left or right depending on the switching delay difference.

The impact of S-TDS where the LEDs are switched according to the approach in Fig. 2 is simulated. We simulated the added delay impairment as a guard delay. This means the guard delay is added once the LED finishes its data transmission, and the other LED transmits an identical copy of data. The S-TDS for Tx_1 is depicted Fig. 4.
We deduce from Fig. 4 that the switching impairment reduces the SINR in the section under Tx_2. In the next time slot, the situation will be reversed, see Fig. 4. We observe that if the IoT node is in the overlapping region, say at 1.5 m; the SINR improves maximum by 9 dB.

Note that we fixed the data rate for all simulations. However, it is expected that if we increase the data rate, the noise power will increase, and the SINR with and without the S-TDS approach will reduce.

In this paper, we presented S-TDS as a simple approach to have SLIPT for multiple coexisted users. Using S-TDS improves the SINR for an unknown IoT position where we harvest light energy opportunistically for green IoT applications. We present the S-TDS for only two LEDs, but if more LEDs (transmitters) are introduced, then a better approach should be studied where the transmitters communicate in groups of non-overlapped regions alternatively.

V. Conclusions and Future Work

This research paper proposed a potential solution for JOE-HC in green IoT networks in healthcare using the VLC system. We provided a complete model for calculating the light irradiation, the harvested energy and the SINR for communication.

In particular, to maximize the performance of JOE-HC, we exploit the usage of switching the LEDs sequentially by

adding a guard delay using the S-TDS approach. This realizes opportunistic energy harvesting for green IoT; meanwhile, improving the SINR in an overlapping region is feasible. The improvement reaches a maximum of 9 dB in the overlapping regions for an optimized guard delay for the switching delay impairments.

In the future, we plan to implement this approach realistically by designing and realising the system's hardware and testing it to deduce its feasibility. We will then compare the S-TDS and other algorithms, such as spatial TDMA, as presented by [21].

Acknowledgment

This research was funded by the Flemish FWO SBO S001521N IoBaLeT project (sustainable Internet of Battery-Less Things).

References

[1] "Cisco Annual Internet Report (2018–2023) White Paper," Mar 2020, Accessed: 28 − 06 − 2022. [Online]. Available: https://www.cisco.com/c/en/us/solutions/collateral/executive-perspectives/annual-internet-report/white-paper-c11-741490.html

[2] J. Ding, M. Nemati, C. Ranaweera, and J. Choi, "IoT Connectivity Technologies and Applications: A Survey," *arXiv preprint arXiv:2002.12646*, 2020.

[3] J. Wu, Y. Qi, G. Gong, J. Fan, M. Miao, W. Yu, J. Ma, and J. L. Drewniak, "Review of the EMC Aspects of Internet of Things," *IEEE Transactions on Electromagnetic Compatibility*, vol. 62, no. 6, pp. 2604–2612, 2020.

[4] T. Wang, F. Yang, J. Song, and Z. Han, "Dimming Techniques of Visible Light Communications for Human-Centric Illumination Networks: State-of-the-Art, Challenges, and Trends," *IEEE Wireless Communications*, vol. 27, no. 4, pp. 88–95, 2020.

[5] J. Hester and J. Sorber, "The Future of Sensing is Batteryless, Intermittent, and Awesome," *Proceedings of the 15th ACM Conference on Embedded Network Sensor Systems*, pp. 1–6, 2017.

[6] K. Hammoud, S. Pollin, and D. Schreurs, "Efficient Indoor Solar Panel Energy Harvesting Exploiting the Crest Factor," in *2022 Wireless Power Week (WPW)*, 2022, pp. 653–657.

[7] S. C. Mukhopadhyay, "Wearable Sensors for Human Activity Monitoring: A Review," *IEEE Sensors Journal*, vol. 15, no. 3, pp. 1321–1330, 2015.

[8] Z. Wang, D. Tsonev, S. Videv, and H. Haas, "Towards Self-Powered Solar Panel Receiver for Optical Wireless Communication," in *2014 IEEE International Conference on Communications (ICC)*, 2014, pp. 3348–3353.

[9] J. Fakidis, H. Helmers, and H. Haas, "Simultaneous Wireless Data and Power Transfer for a 1-Gb/s GaAs VCSEL and Photovoltaic Link," *IEEE Photonics Technology Letters*, vol. 32, no. 19, pp. 1277–1280, 2020.

[10] S. Kadirvelu, W. D. Leon-Salas, X. Fan, J. Kim, B. Peleato, S. Mohammadi, and B. Vijayalakshmi, "A Circuit for Simultaneous Reception of Data and Power Using a Solar Cell," *IEEE Transactions on Green Communications and Networking*, vol. 5, no. 4, pp. 2065–2075, 2021.

[11] J. Kahn and J. Barry, "Wireless Infrared Communications," *Proceedings of the IEEE*, vol. 85, no. 2, pp. 265–298, 1997.

[12] S. Kanwal, J. Rashid, J. Kim, S. Juneja, G. Dhiman, and A. Hussain, "Mitigating the Coexistence Technique in Wireless Body Area Networks By Using Superframe Interleaving," *IETE Journal of Research*, pp. 1–15, 2022.

[13] B. A. A. Modad, E. Yaacoub, A. Kassir, and Z. Dawy, "A Practical TDMA Modification of IEEE 802.11 for Ultra-Dense IoT-Health with Fairness Considerations," in *2020 IEEE International Conference on Communications Workshops (ICC Workshops)*, 2020, pp. 1–6.

[14] L. Fan, X. Liu, H. Zhou, V. C. M. Leung, J. Su, and A. X. Liu, "Efficient Resource Scheduling for Interference Alleviation in Dynamic Coexisting WBANs," *IEEE Transactions on Mobile Computing*, pp. 1–1, 2021.

[15] J. Mu, Y. Wei, H. Ma, and Y. Li, "Spectrum Allocation Scheme for Intelligent Partition Based on Machine Learning for Inter-WBAN Interference," *IEEE Wireless Communications*, vol. 27, no. 5, pp. 32–37, 2020.

[16] R. Ali, B. Kim, S. Kim, H. Kim, and F. Ishmanov, "(ReLBT): A Reinforcement Learning-Enabled Listen before Talk Mechanism for LTE-LAA and Wi-Fi Coexistence in IoT," *Comput. Commun.*, vol. 150, p. 498–505, 2020.

[17] J. Bullough and D. Marcus, "Influence of Flicker Characteristics on Stroboscopic Effects," *Lighting Research & Technology*, vol. 48, no. 7, pp. 857–870, 2016.

[18] W. J. Smith, *Modern Optical Engineering: The Design of Optical Systems*, pp. 219–243, 2000.

[19] N. T. Tong and W. Pora, "A parameter extraction technique exploiting intrinsic properties of solar cells," *Applied Energy*, vol. 176, pp. 104–115, 2016.

[20] I. Moreno, M. A. Alejo, and R. I. Tzonchev, "Designing Light-Emitting Diode Arrays for Uniform Near-Field Irradiance," *Appl. Opt.*, vol. 45, no. 10, pp. 2265–2272, 2006.

[21] J. Beysens, J.-P. M. G. Linnartz, D. Van Wageningen, and S. Pollin, "TDMA Scheduling in Spatially Extended LiFi Networks," *IEEE Open Journal of the Communications Society*, vol. 1, pp. 1524–1538, 2020.